

The Fidelity-Utility Paradox in Surrogate-Based RL for HVAC Control

More accurate surrogate $\rightarrow$ rougher action surface $\rightarrow$ PPO collapse. Hybrid separates rollout smoothness from physical censoring.

Surrogate training environment

Coarse black-box v3

24 h RMSE 1.557 °C · $m_s \approx 0.08$

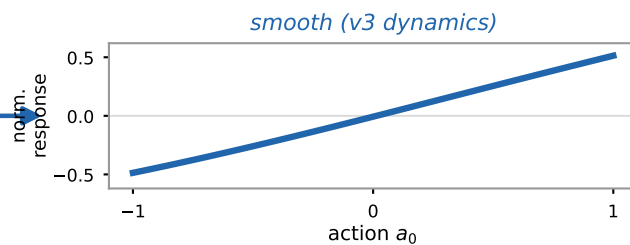
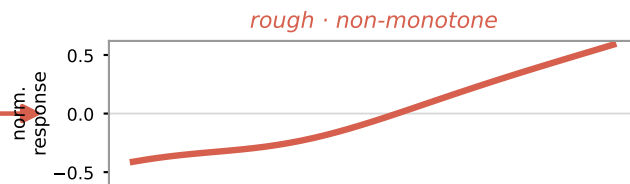
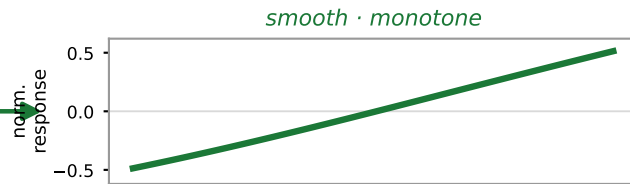
Accurate twin (v3.5 / matched-v3)

RMSE 0.644/0.876 °C · roughness 7.9–9.4 × · $m_s > 1$

Hybrid (v3 rollout + v3.5 censor)

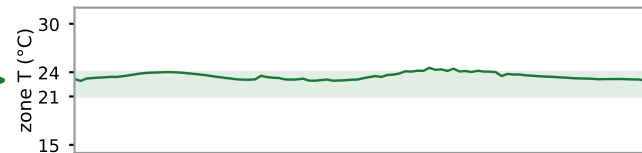
$m_s = 0.041$ (typ.) · violation < 5%

Normalised action $\rightarrow$ next-temperature response

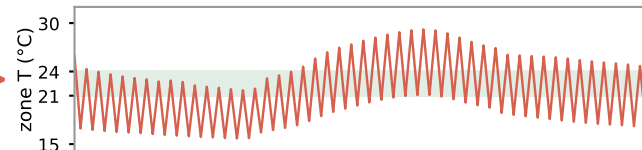


Zone temperature on the live BOPTEST runtime (24 h)

✓ **USABLE**



✗ **COLLAPSE**



✓ **ROBUST**

